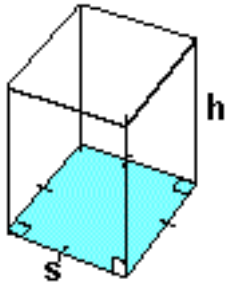


Part 3: Maximize the Volume of a Square-Based Prism

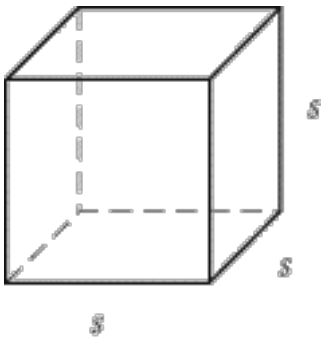
Square-based prism



$$\text{Volume} = s^2 h$$

$$\text{Surface Area} = 2s^2 + 4sh$$

Cube



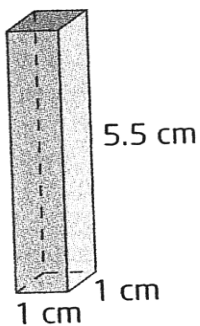
$$\text{Volume} = s^3$$

$$\text{Surface Area} = 6s^2$$

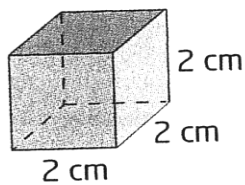
For a given surface area, it is usually optimal to _____ the volume.

Investigation 1: Each of the square-based prisms has a surface area of 24 cm^2 .

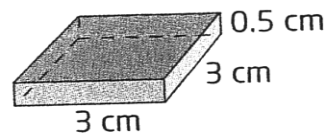
Prism 1



Prism 2



Prism 3

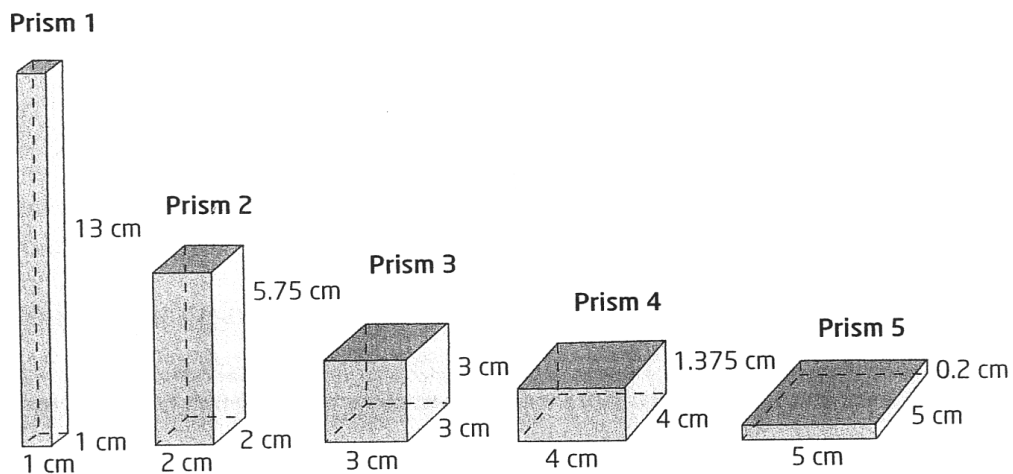


a) Calculate the area of the base and the volume of each prism. Record your data in the following table:

Prism Number	Length of base	Area of base	Surface Area	Height	Volume
1					
2					
3					

b) Which square-based prism has the maximum volume? Describe the shape of this prism compared to the others.

Investigation 2: Each of the square-based prisms has a surface area of 54cm^2 .



a) Predict the dimensions of the square-based prism with maximum volume if the surface area is 54cm^2 .

b) Test your prediction by completing the following table.

Prism Number	Length of base	Area of base	Surface Area	Height	Volume
1					
2					
3					
4					
5					

Results: The maximum volume for a given surface area of a square-based prism always occurs when the prism is a _____.

Example 1: Determine the dimensions of the square-based prism with maximum volume that can be formed using 5400 cm^2 of cardboard.

Solution:

We know that a cube will have the maximum volume

Surface Area of a Cube $= 6s^2$ where s represents the side length

To calculate the side length of the cube given the surface area you must rearrange the formula to isolate s :

$$s = \sqrt{\frac{\text{Surface Area}}{6}}$$

Therefore, if the Surface Area is 5400 cm^2 , $s = \sqrt{\frac{5400}{6}} = 30\text{cm}$. This means that the dimensions of the square-based prism with maximum volume are **30cm x 30cm x 30cm**.

Apply What You Learned

1. Determine the dimensions of the square-based prism with maximum volume for each surface area.

a) 150 cm^2

b) 2400 m^2

c) 750 cm^2

2. Determine the volume of each prism in question 1, to the nearest cubic unit.

a)

b)

c)